

Bhalaji Yadav Kantepalle

Ph.D. Student, Researcher, and Chemical Engineer

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RESEARCH PROFILE

Integrative Life Sciences Ph.D. student and chemical engineer studying how temperature, deformation, and flow change stimuli-responsive soft and liquid-crystalline materials. Uses rheology, quantitative imaging, mechanics, and structure-sensitive characterization to connect processing history to optical and mechanical response.

EDUCATION

Ph.D. Student, Integrative Life Sciences

Virginia Commonwealth University

Aug 2026–Present

Richmond, Virginia

Aug 2026

M.S., Chemical and Life Science Engineering

Virginia Commonwealth University

Richmond, Virginia

- GPA: 3.62
- Thesis: Multi-Criteria Selection of Adhesives for Wearable Textiles
- Advisor: Christina Tang, Ph.D.

Jun 2022

B.Tech., Chemical Engineering

Amrita Vishwa Vidyapeetham

Coimbatore, India

RESEARCH AND TEACHING EXPERIENCE

Graduate Teaching Assistant

Virginia Commonwealth University

Aug 2026–Present

Richmond, Virginia

- Support undergraduate instruction while pursuing doctoral training in the Integrative Life Sciences Ph.D. program.

Sep 2024–Aug 2026

Graduate Researcher, Quantitative Metrology

VCU Soft Functional Materials Lab | Advisor: Christina Tang, Ph.D.

Richmond, Virginia

- Developed mechanics-consistent adhesion and fracture-characterization methods for soft, stretchable, and textile-supported interfaces.
- Engineered reproducible Python and OpenCV workflows that synchronized force-displacement data, deformation tracking, and optical colorimetry.
- Guided two Summer 2026 student research projects in liquid-crystal temperature mapping and hyperspectral detection of plastics in soil mixtures; provided Spring 2026 technical support in NMR characterization, strain-to-color analysis, scientific figures, posters, and grant drafting.
- Contributed technical writing and regulatory analysis to a phased \$100,000 Commonwealth Cyber Initiative proposal; the \$30,000 prototype and proof-of-concept phase was initially funded, with later funding contingent on satisfactory progress.

PEER-REVIEWED PUBLICATIONS

1. Kantepalle, B. Y.; Epitawala Arachchige, U.; Joung, D.; Tang, C. Multi-Criteria Selection of Adhesives for Wearable Textiles. *Polymers* 2026, 18, 1504. <https://doi.org/10.3390/polym18121504>
2. Caloian, I.; Trapp, J.; Kantepalle, B. Y.; Latimer, P.; Lawton, T. J.; Tang, C. Mechanical Properties of Dual-Layer Electrospun Fiber Mats. *Polymers* 2025, 17, 1777. <https://doi.org/10.3390/polym17131777>

RESEARCH SOFTWARE

Peel Trace Evaluation for Soft Substrates

v1.4.0-rc15

Bhalaji Yadav Kantepalle and Christina Tang

Open-source Jupyter and Colab workflow for protocol-defined analysis of force-displacement peel traces from soft substrates. [Repository](#) | [Archived release](#)

PRESENTATIONS, SEMINAR, AND DEFENSE

1. **M.S. Thesis Defense.** B. Y. Kantepalle, "Multi-Criteria Selection of Adhesives for Wearable Textiles,"
Virginia Commonwealth University, Richmond, Virginia. Jul 1, 2026

- Poster Presentation.** B. Y. Kantepalle, U. E. Arachchige, D. Joung, and C. Tang, "Multi-Criteria Selection of Adhesives for Wearable Textiles," PMDC 4th Annual Research Symposium, Virginia Commonwealth University, Richmond, Virginia. Jun 12, 2026
- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," 29th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 22, 2026
- Departmental Research Seminar.** B. Y. Kantepalle, "Beyond Peel Strength: Stability-Driven Adhesion Metrics for Soft Wearable Interfaces," Chemical and Life Science Engineering Research Day, Virginia Commonwealth University, Richmond, Virginia. Apr 15, 2026
- Poster Presentation.** B. Y. Kantepalle, A. Reyna, and C. Tang, "Quantifying Mechanochromic Response in Smart Textiles: A Computer Vision Approach," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 25, 2026
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," ACS Fall 2025, Washington, DC. Aug 20, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," 28th VCU Graduate Student Research Symposium, Richmond, Virginia. Apr 23, 2025
- Poster Presentation.** B. Y. Kantepalle, H. Stwodah, D. Joung, and C. Tang, "Adhesives for Personalized Wearable Devices," VCU Engineering Graduate Research and Postdoc Showcase, Richmond, Virginia. Mar 26, 2025

TEACHING, REVIEW, AND APPLIED INNOVATION

- Classroom Instructor, Undergraduate Material Balance** Nov 11–12, 2025
Virginia Commonwealth University
- Judge, Student Poster Session** Mar 29, 2026
AIChE Atlantic Regional Conference, Virginia Commonwealth University
- Judge, Chem-E-Car Poster Competition** Mar 28, 2026
AIChE Mid-Atlantic Regional Conference, Virginia Commonwealth University
- Prototype Demonstrator, Feedback Friday** Mar 20, 2026
Shelfie Program, VCU da Vinci Center
- Student Volunteer** Feb 2020
Anokha, Amrita University Techfest

INDUSTRY EXPERIENCE

- Technical Project Manager** May 2023–May 2024
Kreative Organics Pvt. Ltd. Hyderabad, India
- Coordinated SAP requalification across quality, engineering, senior management, and external consultants; prepared or supported user requirements, Performance Qualification, gap and impact assessments, and corrective-action documentation.
 - Maintained traceability between process requirements, test evidence, and data-integrity controls.
- Technical Project Management Intern** Oct 2022–Apr 2023
Deep Thought Edu Tech Ventures · placement at Kreative Organics Pvt. Ltd. Hyderabad, India
- Directed a Python chemical-name normalization workflow and built Tableau analyses that reduced recurring trade-data analysis time by about 70%.
 - Co-developed a Notion project-management framework for initiation, tracking, risks, escalation, and completion.

HONORS AND PROFESSIONAL DEVELOPMENT

- Design credentials: [Design Thinking](#) | [Prototyping](#) | [Pitching & Storytelling](#), VCU da Vinci Center, Mar 2026
- Graduate Assistantship Award, Virginia Commonwealth University, Fall 2025
- Selected Participant, 1st National Neutron Scattering School, Oak Ridge National Laboratory, Sep 2025
- [Presentation of Research–Foundational](#), Virginia Commonwealth University, Apr 23, 2025

TECHNICAL SKILLS

Materials and Testing: Adhesion testing, T-peel, fracture mechanics, tensile testing, rheology, viscoelasticity, soft interfaces, wearable textiles

Computational: Python, OpenCV, NumPy, Pandas, SciPy, Matplotlib, image processing, CIE L*a*b* colorimetry, automated analysis

Instrumentation: Universal testing machine, optical microscopy, UV-Vis spectroscopy, Abbe refractometry, 3D printing

Compliance and Process: Validation documentation, data integrity, cGMP, V-model, RCA, CAPA, DOE, Lean manufacturing